Models and Object Classes

VisionAssist AI — Object Detection and Navigation Assistance for Visually Impaired Individuals

Every model file, class name and figure below was read directly from the running project on 9 September 2026 — from the trained weights, `data.yaml`, `backend/detector.py`, `backend/navigation/navigation_engine.py` and `scripts/autolabel.py`. Nothing is quoted from documentation or memory.

1. Every model in the system

Stage	Model	Type	Classes	Why this one
Detection (primary)	`best.pt` YOLO11n fine-tuned 2,591,595 params 5.5 MB	TRAINED	9 footpath classes (§2)	Single-stage, so boxes and classes come out of one forward pass — a two-stage detector cannot meet the latency a walking pace demands. The *nano* size is forced by 4 GB of VRAM shared with the display and with a second model. Measured 19.0 ms per frame.
Detection (indoor)	`yolo11n.pt` COCO-pretrained	PRE-TRAINED	12 indoor classes (§3)	The thesis dataset is Dhaka footpaths and holds almost no indoor imagery, so indoor classes could not honestly be trained from it. Rather than claim a capability the data cannot support, a COCO model supplies furniture only.
Annotation (offline)	`yolov8x-worldv2.pt` YOLO-World, open-vocabulary	OFFLINE	35 text prompts → 9 classes (§5)	The raw pool of 21,359 images had zero annotations, the step at which this project had stalled before. An open-vocabulary detector proposes boxes from plain-language prompts, which a human then reviews. **It never runs at inference time.**
Speech (Bengali)	`bn-BD-NabanitaNeural` via edge-tts	NEURAL TTS	—	Windows SAPI5 on this machine has no Bengali voice installed, so the offline engine physically cannot speak Bengali. This is free, needs no API key, and is the best Bengali quality available. Requires internet.
Speech (English)	`pyttsx3` → Windows SAPI5	SYSTEM TTS	—	Offline and instant. No round trip, so an English warning is never delayed by the network.

Three further Bengali neural voices are configured as alternates and can be selected without a code change: `bn-BD-PradeepNeural` (male, Bangladesh), `bn-IN-BashkarNeural` and `bn-IN-TanishaaNeural` (India). The Bangladesh voice is the default because the users are Bangladeshi.

> **No model performs navigation.** Deciding which obstacle matters and whether to speak is a deterministic rule engine, not a network. It needs no training data, fails in ways that can be enumerated in advance, and can be defended line by line in a viva — all three of which matter more than sophistication in assistive software.

2. The nine trained classes

These are the classes the thesis model was actually trained on. The list was fixed before annotation began and never reordered, because the class index is baked into every label file: reordering would silently relabel the whole dataset.

The rule that produced this list: every class must map to an action the navigation layer can express in a sentence. Nothing decorative was included. A footpath contains hundreds of nameable things — shop signs, litter, parked bicycles' shadows — but a blind pedestrian only needs the ones that change where the next step goes.

ID	Class	Bengali	What it covers	Why it is in the list
0	`person`	মানুষ	Pedestrians, standing or walking	Collision risk, and the dominant obstacle in a Dhaka crowd.
1	`pole`	খুঁটি	Electric, utility and sign poles; guy-wire anchor posts planted mid-path	The single most common fixed obstruction in the middle of a Dhaka footpath.
2	`tree`	গাছ	Trunks growing through the walkway	Head and body collision; a cane sweeping low can miss a trunk at chest height.
3	`stairs`	সিঁড়ি	Steps up or down, kerb ramps, shop entrance steps	Fall risk. Needs a distinct warning because the correct response is to stop, not to step around.
4	`pothole`	গর্ত	Broken or missing pavement, sunken slabs, rubble craters	Trip and ankle injury. Invisible to a cane until contact.
5	`open_drain`	খোলা ড্রেন	Uncovered drains, missing or displaced manhole covers	The most dangerous class in the set: a fall into an open drain is a serious injury, and covers go missing often enough in Dhaka that this is routine, not hypothetical.
6	`parked_vehicle`	পার্ক করা গাড়ি	Rickshaw, CNG auto, motorcycle, car or van parked on or across the footpath	Blocks the walkway entirely and forces a step into traffic.
7	`street_obstacle`	রাস্তার প্রতিবন্ধকতা	Vendor stalls, construction material, bricks, sand piles, bins, hanging signboards	Forces a detour. Deliberately one broad class: the required action is identical for all of them, so splitting it would add annotation cost without changing any instruction.
8	`curb`	ফুটপাতের কিনারা	The footpath edge where it drops to the road	The boundary between safe surface and live traffic — arguably the single most important line to stay on the correct side of.

2.1 How much data each class actually got, and what it achieved

Class	Train	Val	Test	AP@0.5	Recall	Verdict
`parked_vehicle`	9,290	2,652	1,360	0.781	0.688	Reliable
`person`	6,178	1,750	878	0.837	0.749	Reliable
`street_obstacle`	549	183	99	0.295	0.267	Weak
`tree`	422	118	65	0.237	0.262	Weak
`curb`	144	35	20	0.359	0.450	Weak
`pole`	109	35	9	0.268	0.222	Too few to judge
`stairs`	17	4	2	0.995 *	1.000	Noise, not evidence
`pothole`	17	3	1	0.000	0.000	Not detected
`open_drain`	12	9	1	0.000	0.000	Not detected

Read this table before quoting any class result. Per-class AP tracks instance count almost monotonically, which says the binding constraint is annotation coverage — not model capacity and not training length.

* `stairs` at 0.995 is computed over *two* test boxes. That is statistical noise. `pothole` and `open_drain` show zero recall: the model predicts neither class at all, and their precision of 1.000 is the convention used when no positive prediction was ever made, not a sign of quality.

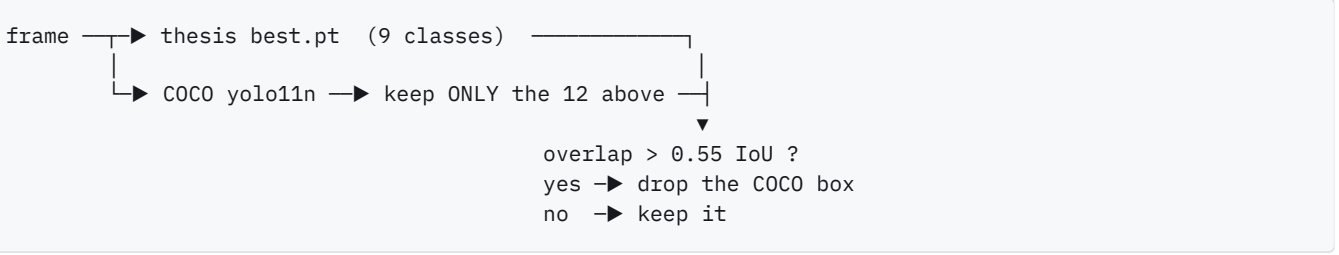
The consequence must be stated plainly: the system is reliable on people and vehicles and is *blind to the three fall hazards* that motivated the class list. Section 5 explains why those classes were starved of data.

3. The twelve indoor classes

Supplied at runtime by the COCO-pretrained model, never by the thesis model. The fusion rule is in `backend/detector.py`.

Class	Bengali	Why it is included
`chair`	চেয়ার	The commonest indoor trip hazard; legs are low and a cane can pass between them.
`dining table`	টেবিল	Waist-height edge, and the class COCO uses for tables generally.
`bed`	বিছানা	Large low obstacle that defines the walkable area of a room.
`couch`	সোফা	As above; distinguished from `bed` by COCO, so kept separate.
`bench`	বেঞ্চ	Appears indoors and on footpaths; useful in both.
`potted plant`	গাছের টব	Frequently placed in corridors and doorways.
`tv`	টেলিভিশন	Wall- or stand-mounted at head height.
`refrigerator`	ফ্রিজ	Full-height fixed obstacle; a useful landmark for orientation.
`oven`	ওভেন	Protruding appliance at waist height.
`microwave`	মাইক্রোওয়েভ	Counter-height protrusion.
`sink`	সিংক	Fixed waist-height obstacle; orients a user in a kitchen or washroom.
`toilet`	টয়লেট	Orientation landmark in an unfamiliar washroom.

3.1 The two rules that keep the fusion honest



1. **The indoor model is excluded from every class the thesis model covers** — notably `person`, which both can detect. Without this the same pedestrian would be listed and announced twice.
2. **On any overlap above 0.55 IoU the thesis model wins**, because it is the model trained on this thesis's data.

> **Declare this in the defence.** These twelve classes are COCO-pretrained and carry **no thesis-measured mAP**. They are a deployment convenience, not a research contribution.
>
> `door` is deliberately absent. COCO has no door class and the thesis dataset has no door annotations, so supporting it would mean inventing a capability that was never measured.

4. How the classes are ranked

Not every class carries equal weight. The navigation engine sorts detections by a score whose terms are readable rather than tuned, so any ordering can be explained.

Tier	Score	Classes	Reasoning
DANGER	+100	`pothole` `open_drain` `stairs` `curb` `pole` `car` `bus` `truck` `motorcycle` `bicycle` `train`	Fall or vehicle-collision hazards. A mistake here injures.
INDOOR	+40	The twelve furniture classes of §3	Worth announcing, but walking into a chair is not walking into an open drain.
OTHER	0	`person` , `tree` , `street_obstacle` , `parked_vehicle` and remaining COCO classes	Announced on geometry alone. A person is usually avoidable and often moves.
Geometry		CENTER +50 · NEAR +30 / MEDIUM +12 / FAR 0 · confidence × 10	Position and proximity dominate confidence, so the detector's certainty can never outrank being directly in the path.

A further **31 COCO classes are permanently ignored** and never announced — food, sports equipment, small handheld items such as `toothbrush` , `book` and `cell phone` . They are real detections but nothing a pedestrian navigates around, and narrating them would bury the warnings that matter.

5. Prompts used to annotate each class

YOLO-World was driven with several descriptive phrases per class rather than the bare class name; open-vocabulary detectors respond markedly better to a description than to a label, and several phrasings recover objects any single phrasing misses. Thirty-five prompts map back onto the nine classes.

Class	Prompts	Boxes found
`person`	person · pedestrian walking · man standing · woman walking	8,820
`pole`	utility pole · electric pole on footpath · lamp post · sign post	153
`tree`	tree trunk · tree growing on pavement	606
`stairs`	stairs · concrete steps · staircase entrance	23
`pothole`	pothole · broken pavement · hole in the road surface · damaged footpath tiles	21
`open_drain`	open drain · uncovered drain · manhole without cover · open sewer gap in pavement	22
`parked_vehicle`	parked car · rickshaw · auto rickshaw · parked motorcycle · parked van · bicycle parked on footpath	13,328
`street_obstacle`	street vendor stall · pile of bricks · construction material · garbage bin · sand pile on footpath · signboard on pavement	833
`curb`	kerb edge of footpath · raised pavement edge	204

This table is the root cause of the failures in §2.1. Three classes — `pothole`, `open_drain` and `curb` — are flagged in the code as `LOW_TRUST` because the annotation model cannot find them.

A pothole has no consistent visual prototype: it is defined by the *absence* of intact surface, not by a characteristic shape, and is easily confused with a shadow, a repair patch or an ordinary paving seam. The open-vocabulary model therefore proposed almost no candidates, the human review had almost nothing to confirm, and those classes reached training with 12–17 instances.

The fix is annotation, not architecture: hand-draw 100–200 instances of each hazard class and retrain. The images already exist in the unused pool.

6. Summary

Question	Answer
How many models run at inference?	**Two** detectors (thesis `best.pt` + COCO `yolo11n.pt`) and one text-to-speech voice per language.
How many classes can be detected?	**21** — 9 trained on thesis data, 12 borrowed from COCO.
How many are thesis contributions?	**9.** Only these carry measured mAP.
How many work reliably today?	**2** (`person`, `parked_vehicle`). Four are weak, three are not detected at all.
Which model is *not* in the running system?	`yolov8x-worldv2.pt` — annotation only, offline.
Which component uses no model?	Navigation, spatial reasoning and sentence generation — all deterministic rules.